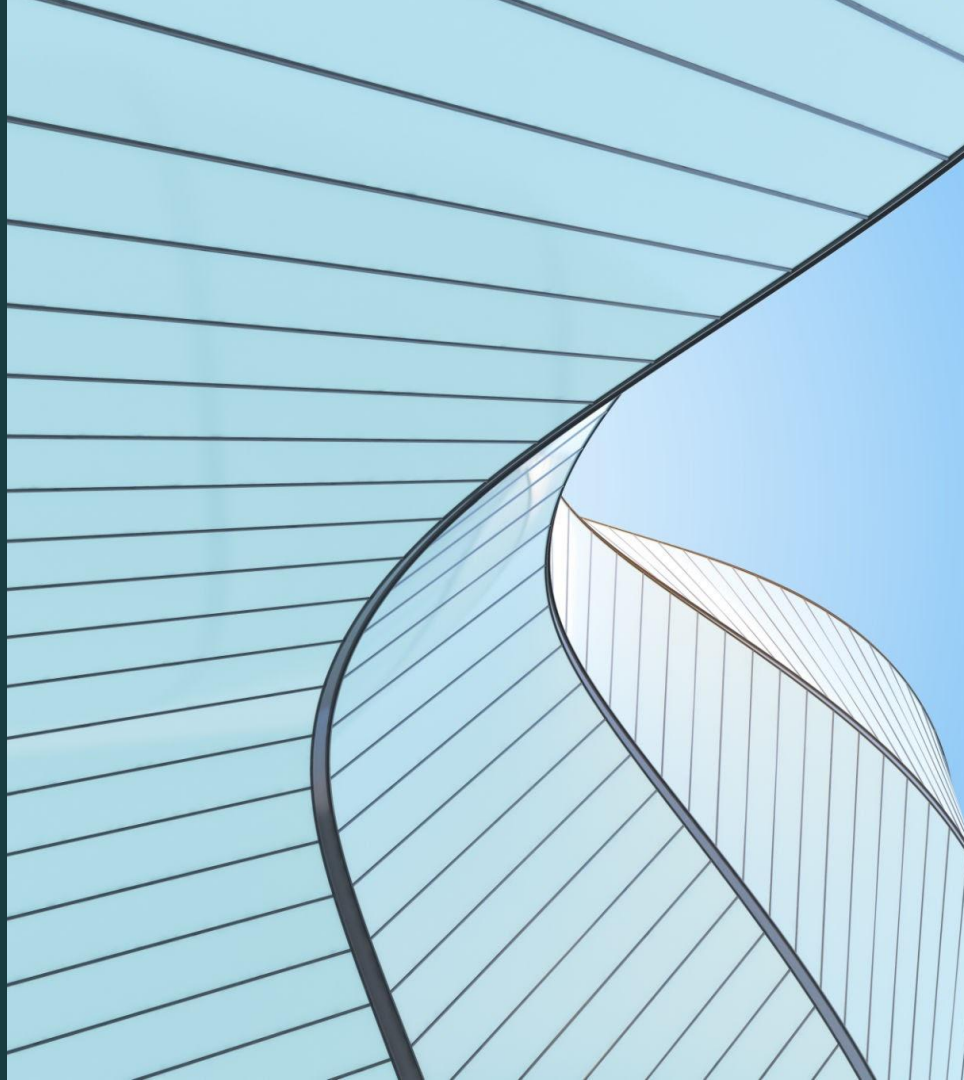
Cloud Native Data Engineering
Assignment 1

Campus Energy Monitoring Platform

Cloud system design and pricing on AWS
Domain: Campus IoT / Smart Building

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Business Context

A campus facilities team runs 124 buildings and 308 hourly meters at a university campus in Austin, TX.

Facility managers lack real-time visibility, cost mapping to individual buildings, and only discover anomalies once monthly bills arrive.

This leads to wasted energy and missed sustainability targets.

Success Criteria

- Reduce total campus energy consumption by 5% within a year.
- Provide hourly energy dashboards for each building with under 5-minute latency.
- Detect abnormal energy consumption patterns within 24 hours.
- Attribute 100% of energy costs to individual buildings.
- Achieve 95% data completeness across all meters.

Problem Statement

Design a cloud-based data platform that ingests hourly meter readings, building metadata, weather data, and electricity prices to provide dashboards, cost allocation, and anomaly detection for the campus facilities management office..

Stakeholders

Facility Managers: Real-time dashboards and alerts.

Sustainability: Energy and carbon reporting.

Finance/Accounting: Accurate cost allocation per building.

Operations Teams: Secure data ingestion from meters and external APIs.

Solution Overview

Platform Summary

Collection:

Hourly meter readings (electricity, chilled water, steam, hot water), building metadata, weather data, and electricity price data.

Processing:

Ingests batch CSV files and API data, normalizes into a relational model, stores in a data lake and warehouse.

Analysis:

Dashboards for energy consumption, peak demand, cost per square meter, data completeness, and anomaly alerts.

Output:

QuickSight dashboards, monthly cost reports, and SQL-based anomaly detection.

Processing Mode

Primarily Batch / Near-real-time

Justification:

Primarily batch for historical BDG2 CSV data and API polling for EIA price data..

Supports near-real-time streaming via AWS IoT Core and MQTT for future live sensors.

Hybrid approach optimizes for compute costs.

Data Sources

Meter Readings

Source:
BDG2
Bull site meter readings.

Type: Batch
Format: CSV
Frequency: Hourly
Volume: 308 Records/hr
Internal

Building & Meter Metadata

Source:
BDG2
Bull site building metadata.

Type: Batch
Format: CSV
Frequency: Once
Volume: 124 Buildings
Internal

Weather at Site

Source:
BDG2
Bull site weather data.

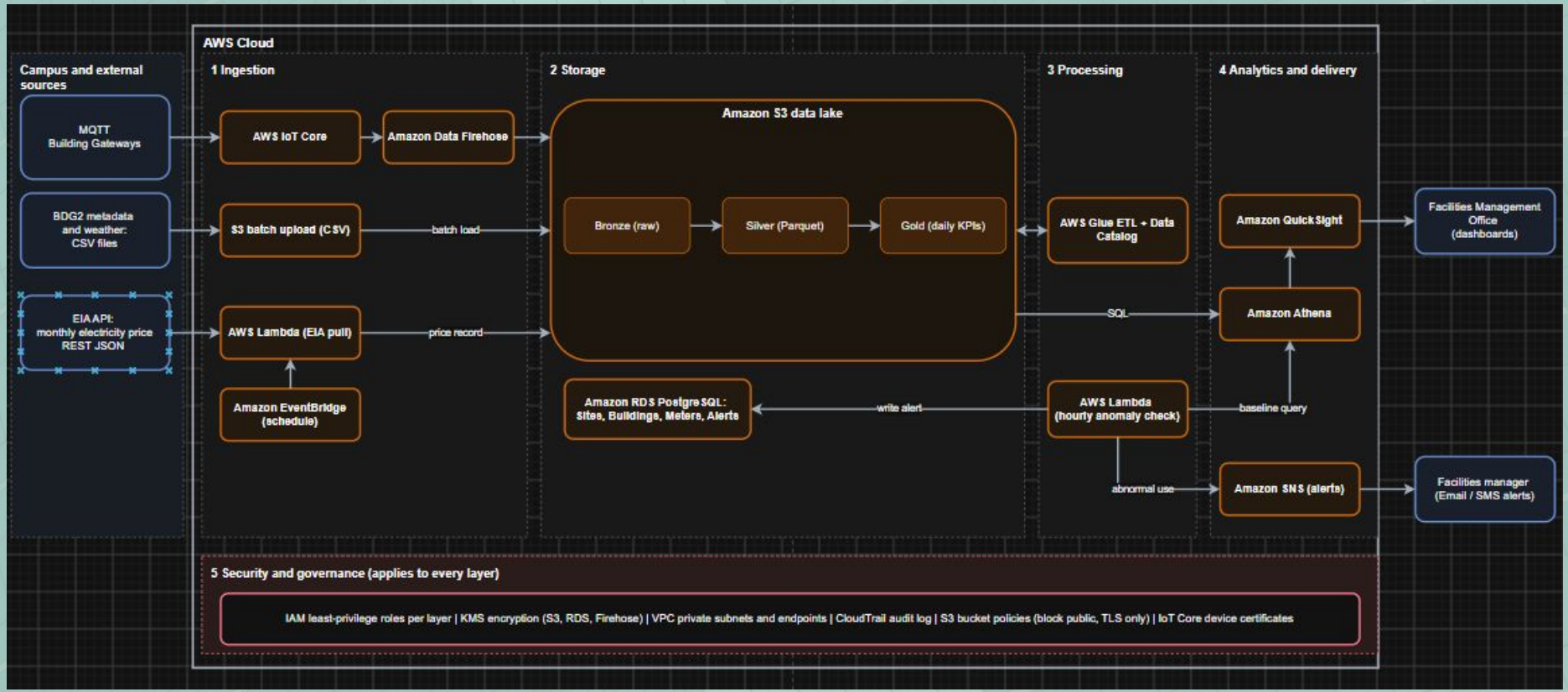
Type: Batch
Format: CSV
Frequency: Hourly
Volume: 1 Record/site
Internal

Electricity Pricing

Source:
EIA
Texas / ERCOT

Type: REST API
Format: JSON
Frequency: Monthly
Volume: 1 Record/region
External

AWS Architecture by Layer



https://drive.google.com/file/d/1QYY5CyCg_tDOr0BLrXmcJnV-UXLua81X/view?usp=sharing

AWS Services Justification

Service	Why chosen	Alternative rejected
AWS IoT Core	Per-gateway certificates, MQTT	Direct HTTPS: no device identity
Amazon Data Firehose	Managed buffered delivery to S3	Custom writer Lambda
Amazon S3	Cheap Bronze/Silver/Gold lake	Warehouse-only storage
Lambda + EventBridge	Scheduled EIA pull, hourly anomaly check	Always-on EC2 cron
AWS Glue	Serverless ETL and catalog	EMR clusters
Amazon Athena	Pay per query, light usage	Redshift (always-on)
Amazon RDS PostgreSQL	Joins across sites, buildings, meters	DynamoDB (no joins)
Amazon QuickSight	AWS-native dashboards	Power BI, outside AWS
Amazon SNS	Alert to email, SMS	Custom mailer
IAM, KMS, VPC, CloudTrail	Access, encryption, isolation, audit	None; baseline controls

Data Sizing & Capacity Planning

Data Stream	Basis	Records per Month	Size	Usage per month
Meter readings	308 * 24 hrs * 30 days =	221,760	0.12 KB	26.6 MB
Weather	Hourly, 1 Site	720	0.5 KB	0.4 MB
EIA API Prices	Monthly	4	1 KB	4 KB

- 12-month S3 growth: Bronze about 0.32 GB, Silver about 0.08 GB (assumed 4:1 Parquet), Gold about 0.02 GB (daily aggregates)
- One-time backfill: 2016–2017 history, 308 × 17,544 = 5.4M rows = 0.65 GB
- Concurrency (assumed): About 15 users, 3–5 simultaneous Athena queries, 5–8 QuickSight sessions, 20–30 RDS connections at peak usage.

Cost Estimation (AWS Pricing Calculator)

Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
Amazon Simple Storage Servi...	-	0.00 USD	1.22 USD	-	US East (Ohio)	S3 Standard storage (1 GB per ...
AWS IoT Core	-	0.00 USD	1.30 USD	-	US East (Ohio)	Number of devices (MQTT) (308...
Amazon Data firehose	-	0.00 USD	0.00 USD	-	US East (Ohio)	Source Type (Direct PUT or Kine...
Amazon Simple Storage Servi...	-	0.00 USD	0.09 USD	-	US East (Ohio)	DT Inbound: All other regions (1...
AWS Lambda	-	0.00 USD	0.28 USD	-	US East (Ohio)	Architecture (x86), Architecture ...
AWS Glue	-	0.00 USD	0.75 USD	-	US East (Ohio)	Number of DPUs for Apache Sp...
Amazon Athena	-	0.00 USD	0.89 USD	-	US East (Ohio)	Total number of queries (308 pe...
Amazon RDS for PostgreSQL	-	0.00 USD	1,359.48 USD	-	US East (Ohio)	Storage amount (100 GB), Stora...
Amazon QuickSight	-	0.00 USD	930.50 USD	-	US East (Ohio)	SPICE capacity in gigabytes (GB)...
Amazon Simple Notification S...	-	0.00 USD	0.00 USD	-	US East (Ohio)	
AWS CloudTrail	-	0.00 USD	0.00 USD	-	US East (Ohio)	Management events units (milli...

Business Value & Cost Optimization

Scenario	Energy Saved	Value
Low	2%	$108K
Baseline	4%	$216K
High	6%	$324K

Value statement: Catching waste earlier, a 4% saving on about $5.4M of electricity is roughly $216K a year (range $108K to $324K), compared with a run cost of $2294.51 a month.

Optimization: store readings as Parquet partitioned by site and date so Athena scans only needed files. Data volume is tiny, so always-on services (RDS, QuickSight) are expected to dominate cost; confirm in the Calculator.